MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS)
THIRD SEMESTER EXAMINATION
MG3DSCMAT203- Essentials of Quantitative Decision Making
(2024 Admission Onwards)

Duration: 2 hrs Maximum Marks: 70

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Interest (I), Appreciation (Ap), and Skill (S)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability

[Learning Domain] [CO No(s)]

Part A
Short Answer Type Questions
Answer any five questions. Each question carries 2 marks

1. A student scored 40, 50, 60, 70, 30 and 60 in six tests. Find the mean and median score of the student? [U] [1]
2. The mean of 100 observation is 50 and the standard deviation is 10. What will be the new mean and S.D if 5 is added to each observation [An] [1]
3. Define Karl Pearson's coefficient of correlation (**r**) and state the range of values of **r** [K] [2]
4. Find byx , if 2x+4y-5 = 0 is the regression equation of y on x [A] [2]
5. Define slack variables in LPP. Explain with an example. [K] [3]
6. What is meant by a feasible solution in LPP [U] [3]
7. What is an unbalanced transportation problem [K] [4]
8. What is meant by degeneracy in a transportation problem [U] [4]

Part B
Short Essay Type Questions
Answer any five questions. Each question carries 6 marks

9. Compute the **mode** of the given frequency distribution [A] [1]

Class	0-10	10-20	20-30	30-40	40-50	50-60	60-70	70-80	80-90
Frequency	3	8	12	7	25	23	14	5	1

10. Calculate the Mean Deviation about the mean of the following data. [A] [1]

Marks	5	8	10	11	15
Frequency	4	6	8	7	5

11. Find Karl Pearson's correlation coefficient for the following data. Also, interpret the result. [A_n] [2]

Price (X)	5	10	12	15	18	25
Demand(Y)	25	22	18	12	10	12

12. Find the regression equation of y on x [A] [2]

X	2	4	6	8	10
Y	5	7	9	8	11

13. A factory produces two types of toys: A and B. Each unit of Toy A requires 3 hours of machining and 2 hours of painting. Each unit of Toy B requires 2 hours of machining and 1 hour of painting. The factory has a total of 120 hours of machining and 70 hours of painting available. The profit is ₹30 for each Toy A and ₹20 for each Toy B. Formulate the LPP to maximize the profit. [A_n] [3]

14. Write the standard form of the LPP [U] [3]

Maximize: $Z = 30x + 20y - 10z$

Subject to:

$3x+2y+2Z \leq 120$

$2x+y-3Z \geq 70$

$3x-2y \geq -10$

$x, y, z \geq 0$

15. Obtain an initial basic feasible solution to the following transportation problem using the **NWC Rule**. [A] [4]

	D1	D2	D3	Capacity
O1	1	2	4	60
O2	4	3	2	40
O3	0	2	1	30
Demand	40	50	10	

16. Find the initial basic feasible solution of the following transportation problem using **VAM** method [A] [4]

	A	B	C	Supply
P	4	8	8	50
Q	12	8	11	60
R	10	6	3	40
Demand	20	80	50	

Part C
Essay Type Questions
Answer any three questions. Each question carries 10 marks

17. In a test given to two groups of students the scores obtained are as follows: [A_n] [1]

Group1	23	11	19	26	35	46	53	18	36
Group2	31	18	21	31	48	40	18	23	30

Which group is more consistent and which group is more intelligent

18.Ten competitors in a beauty contest are ranked by three judges in the following order: [A] [2]

Competitors	1	2	3	4	5	6	7	8	9	10
1st judge	1	6	5	10	3	2	4	9	7	8
2nd judge	3	5	8	4	7	10	2	1	6	9
3rd judge	6	4	9	8	1	2	3	10	5	7

Use the rank correlation coefficient to discuss which pair of judges have the nearest approach to common taste in beauty

19. In a partially destroyed record of a correlation analysis, the following regression equations are available: $8x-10y+66=0$ and $40x-18y=214$ Find (i) The mean values of x and y (ii) the coefficient of correlation **r** [A] [2]

20. Solve the following LPP using graphical method:- [U] [3]

Minimize $Z = X_1 + X_2$

subject to $2X_1 + X_2 \geq 4$

$X_1 + 7X_2 \geq 7$

$X_1, X_2 \geq 0$

21.Solve by Simplex method. [A] [3]

Maximize $Z= 40x+30y$

Subject to

$x+y \leq 12$

$2x+y \leq 16$

$x, y \geq 0$

22.A company has three factories located in three cities viz. X, Y, Z. These factories supplies consignments to four dealers viz. A, B, C and D. The dealers are spread all over the country. The production capacity of these factories are 310, 100 and 290 units per month respectively. The net return per unit product is given in the following table. [A] [4]

Factory/Dealers	A	B	C	D	Capacity
X	6	6	6	4	310
Y	4	2	4	5	100
Z	5	6	7	8	290
Requirement	150	130	120	300	

Determine a suitable allocation to **maximize** the total return.